

Algorithmic Trading and Quantitative Strategies

Part 7

Dr. Ayhan Yuksel, CFA, FRM, PRM

- 1 Foundations of Factor Investing
- 2 Selected Factors
- 3 Popularity of Factor Investing

Foundations of Factor Investing

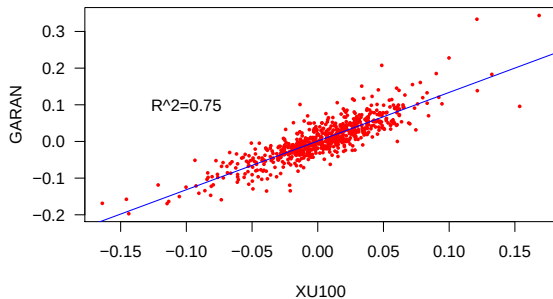
The Most Fundamental Question of Asset Pricing

What drives asset returns?

Market Factor

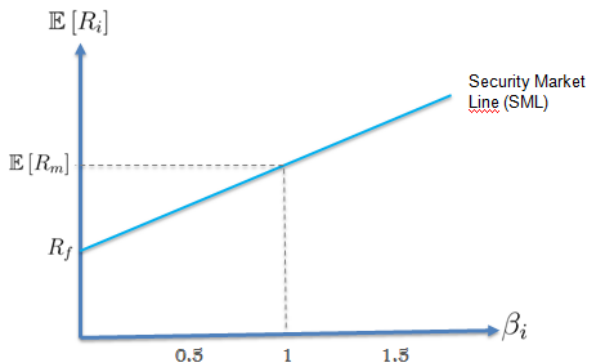
$$R_i - R_f = \alpha_i + \beta_i(R_m - R_f) + \epsilon_i$$

Weekly Returns



Capital Asset Pricing Model

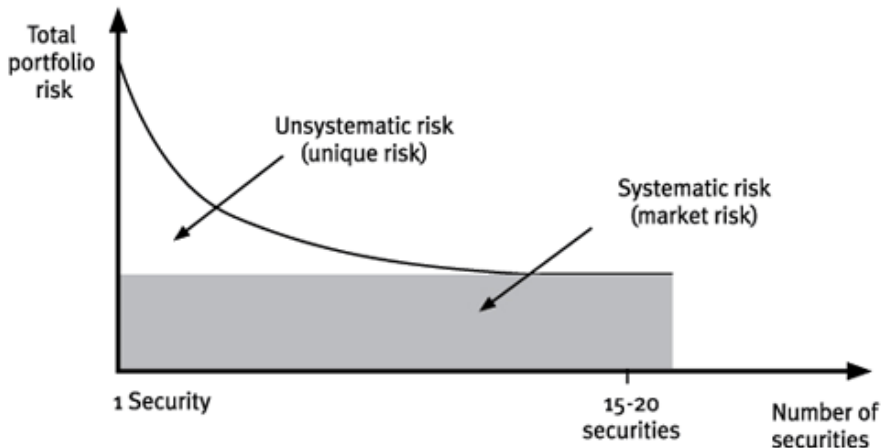
$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_m] - R_f)$$



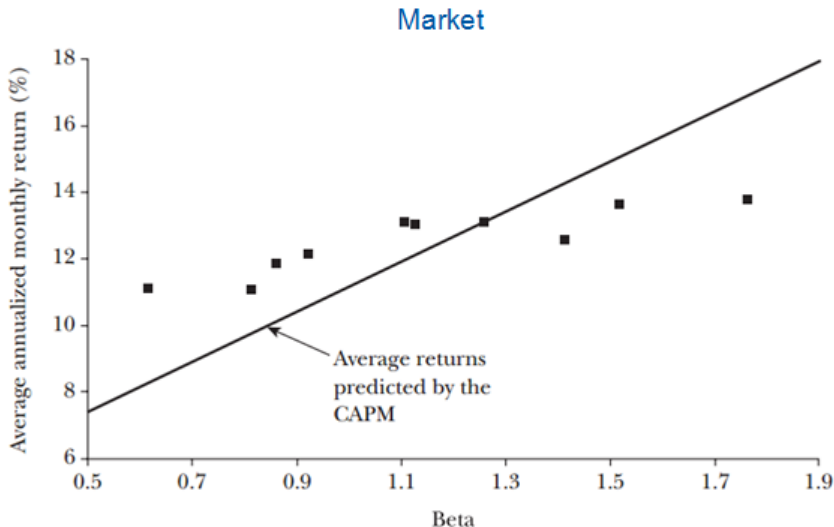
Capital Asset Pricing Model

- The oldest and most well-known model of stock returns is the Capital Asset Pricing Model (CAPM) - CAPM became a foundation of modern financial theory in the 1960s
- In the CAPM, securities have only two main drivers: systematic risk and idiosyncratic risk.
- Systematic risk in the CAPM is the risk that arises from exposure to the market and is captured by beta, the sensitivity of a security's return to the market.
- Since systematic risk cannot be diversified away, investors are compensated with returns for bearing this risk.
- In other words, the expected return to any stock could be viewed as a function of its beta to the market.

Capital Asset Pricing Model



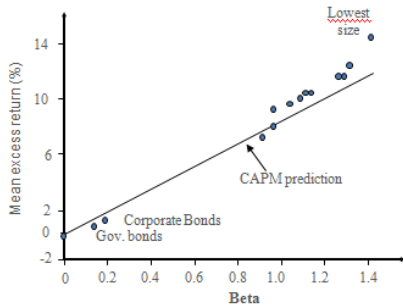
Capital Asset Pricing Model



Source: Fama and French (2004)

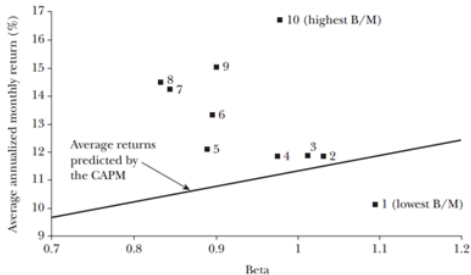
Capital Asset Pricing Model

Size



Source: Cochrane (2000)

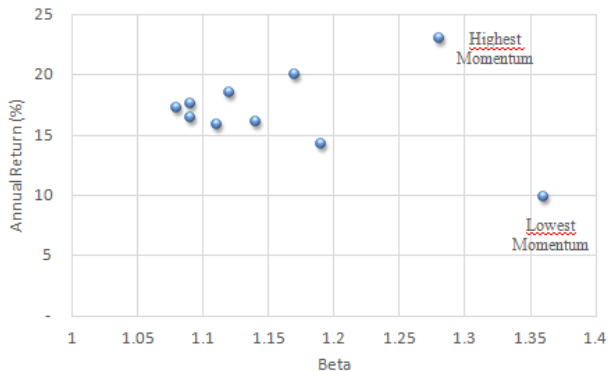
Value



Source: Fama and French (2004)

Capital Asset Pricing Model

Momentum



Source: Jegadeesh and Titman (1993)

Factor Models

- Chaos

$$\mathbb{E}[R_i] - R_f = \boxed{\alpha_i}$$

- Sharpe (1964)

$$\mathbb{E}[R_i] - R_f = \beta_i^m (\mathbb{E}[R_m] - R_f)$$

- Chaos again

$$\mathbb{E}[R_i] - R_f = \boxed{\alpha_i} + \beta_i^m (\mathbb{E}[R_m] - R_f)$$

- Fama and French (1992)

$$\mathbb{E}[R_i] - R_f = \beta_i^m (\mathbb{E}[R_m] - R_f) + \beta_i^{smb} \mathbb{E}[R_{smb}] + \beta_i^{hml} \mathbb{E}[R_{hml}]$$

Factor Models

- Chaos again

$$\mathbb{E}[R_i] - R_f = \boxed{\alpha_i} + \beta_i^m (\mathbb{E}[R_m] - R_f) + \beta_i^{smb} \mathbb{E}[R_{smb}] + \beta_i^{hml} \mathbb{E}[R_{hml}]$$

- Carhart (1997)

$$\mathbb{E}[R_i] - R_f = \beta_i^m (\mathbb{E}[R_m] - R_f) + \beta_i^{smb} \mathbb{E}[R_{smb}] + \beta_i^{hml} \mathbb{E}[R_{hml}] + \beta_i^{wml} \mathbb{E}[R_{wml}]$$

- Chaos again

$$\begin{aligned} \mathbb{E}[R_i] - R_f &= \boxed{\alpha_i} + \beta_i^m (\mathbb{E}[R_m] - R_f) + \beta_i^{smb} \mathbb{E}[R_{smb}] + \\ &\quad \beta_i^{hml} \mathbb{E}[R_{hml}] + \beta_i^{wml} \mathbb{E}[R_{wml}] \end{aligned}$$

- Etc.

Factor Models

CAPM:

$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_m] - R_f)$$



Fama-French:

$$\mathbb{E}[R_i] - R_f = \beta_i^m(\mathbb{E}[R_m] - R_f) + \beta_i^{smb}\mathbb{E}[R_{smb}] + \beta_i^{hml}\mathbb{E}[R_{hml}]$$



Arbitrage Pricing Theory (APT):

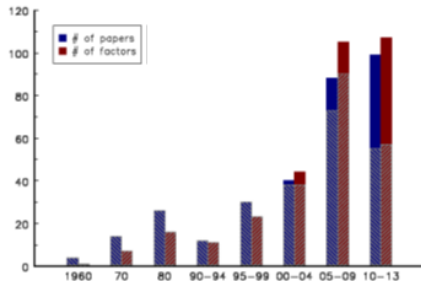
$$\mathbb{E}[R_i] - R_f = \sum \beta_i^k(\mathbb{E}[F_k] - R_f)$$

Factor Models

- In general, a factor can be thought of as any characteristic relating a group of securities that is important in explaining their returns and risk.
- As noted in the early CAPM-related literature, the market can be viewed as the first and most important equity factor.
- Researchers generally look for factors that are persistent over time and have strong explanatory power over a broad range of stocks.
- Since, unlike stock returns, factors cannot be directly observed, there of course remains a vigorous debate about how to define and estimate them.
- There are three main categories of factors today: macroeconomic, statistical, and fundamental.
- Macroeconomic factors include measures such as surprises in inflation, surprises in GNP, surprises in the yield curve, and other measures of the macro economy.
- Statistical factor models identify factors using statistical techniques such as principal components analysis (PCA) where the factors are not pre-specified in advance.
- Arguably the mostly widely used factors today are fundamental factors. Fundamental factors capture stock characteristics such as industry membership, country membership, valuation ratios, and technical indicators, to name a few.

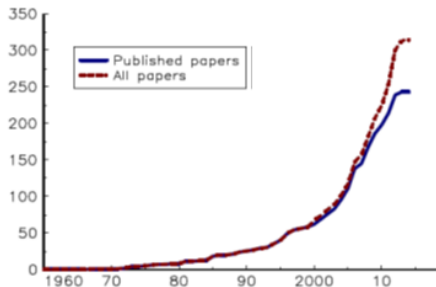
Factor Models

Number of New Factors



Source: Cazalet and Roncalli (2014)

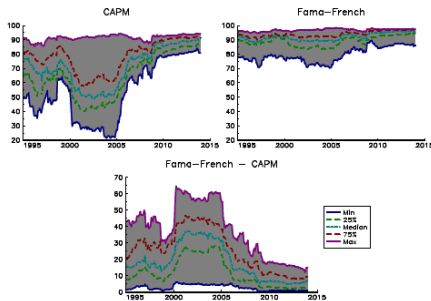
Cumulative Number of Factors



Explanatory Power of Factor Models

Cross-section = 100 value-weighted portfolios (independent sorts into 10 size groups and 10 B/M groups)

Figure: R^2 coefficient (in %) – US

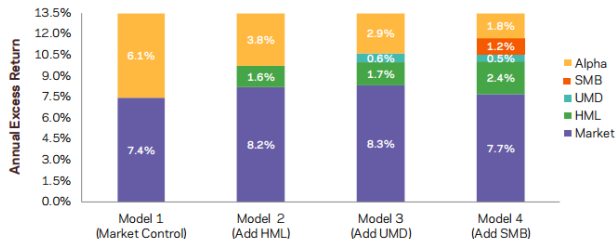


Factor Models

Part A: Regression Results

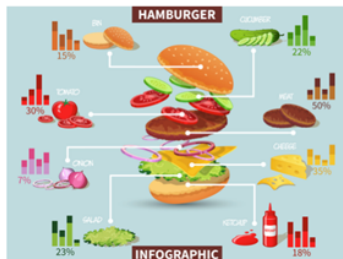
	Model 1 (Market Control)	Model 2 (Add HML)	Model 3 (Add UMD)	Model 4 (Add SMB)
Alpha (ann.)	6.1%	3.8%	2.9%	1.8%
t-statistic	3.6	2.5	1.9	2.2
Market Beta	0.96	1.05	1.07	0.99
t-statistic	31.1	35.7	36.0	61.5
HML Beta		0.43	0.46	0.65
t-statistic		9.8	10.3	26.4
UMD Beta			0.09	0.07
t-statistic			3.0	4.6
SMB Beta				0.74
t-statistic				32.2
R ²	0.70	0.75	0.76	0.93

Part B: Portfolio Return Decomposition



Factor Models

Traditional View



Factor View



Stock	Weight
GARAN	25%
EKGYO	10%
TUPRS	15%
SISE	10%
THYAO	30%
KCHOL	10%

Factor	Weight
Value	15%
Size	5%
Momentum	25%
Sentiment	20%
Profitability	15%
Growth	20%

Selected Factors

Factors

- Macroeconomic Factors
 - Surprises in interest rates / inflation
 - Surprises in gross domestic production
 - Surprises in industrial production
- Fundamental Factors (Cross-Sectional Characteristics)
 - Market Value / Book Value
 - Return on Equity
 - Past Return
 - Dividend Yield
- Statistical/Latent Factors
 - Principal Component Analysis
 - Exploratory Factor Analysis

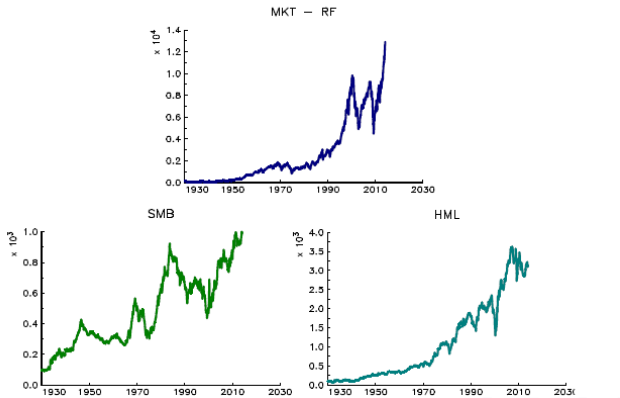
Factors

Exhibit 1: Well-Known Systematic Factors from the Academic Research

Systematic Factors	What It Is	Commonly Captured by
Value	➤ Captures excess returns to stocks that have low prices relative to their fundamental value	➤ Book to price, earnings to price, book value, sales, earnings, cash earnings, net profit, dividends, cash flow
Low Size (Small Cap)	➤ Captures excess returns of smaller firms (by market capitalization) relative to their larger counterparts	➤ Market capitalization (full or free float)
Momentum	➤ Reflects excess returns to stocks with stronger past performance	➤ Relative returns (3-mth, 6-mth, 12-mth, sometimes with last 1 mth excluded), historical alpha
Low Volatility	➤ Captures excess returns to stocks with lower than average volatility, beta, and/or idiosyncratic risk	➤ Standard deviation (1-yr, 2-yrs, 3-yrs), Downside standard deviation, standard deviation of idiosyncratic returns, Beta
Dividend Yield	➤ Captures excess returns to stocks that have higher-than-average dividend yields	➤ Dividend yield
Quality	➤ Captures excess returns to stocks that are characterized by low debt, stable earnings growth, and other "quality" metrics	➤ ROE, earnings stability, dividend growth stability, strength of balance sheet, financial leverage, accounting policies, strength of management, accruals, cash flows

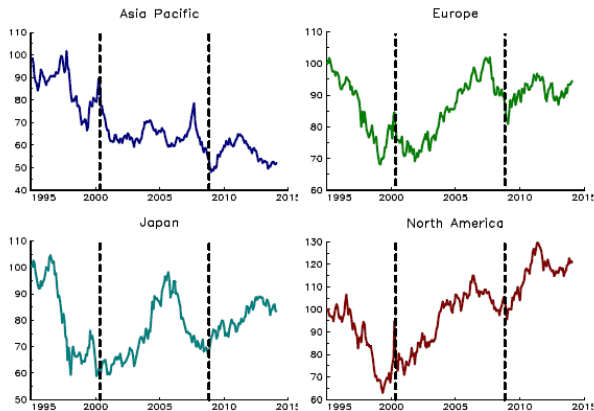
Fama-French Model

Figure: Fama-French US risk factors (1930 - 2013)



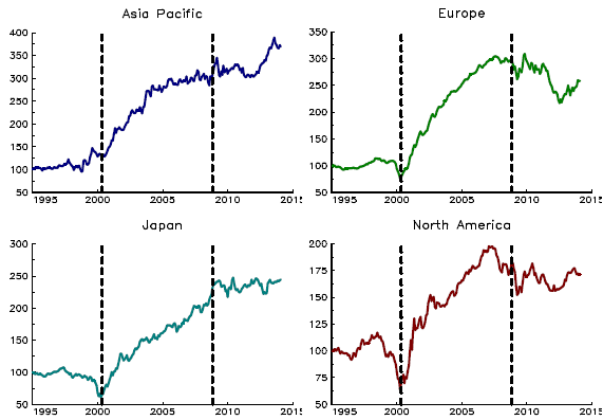
Fama-French Model

Figure: Fama-French SMB factor (1995 – 2013)



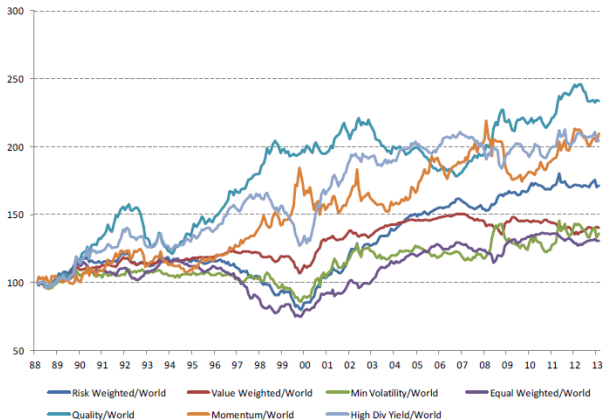
Fama-French Model

Figure: Fama-French HML factor (1995 – 2013)



Is Factor Investing a Free Lunch

Exhibit 7: All Systematic Factors are Cyclical (Cumulative Relative Returns, June 1988 to June 2013)



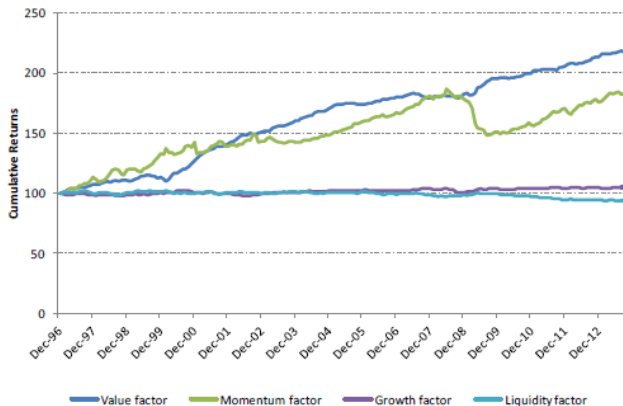
Is Factor Investing a Free Lunch

How to Reduce Factor Drawdowns?

- Setting an appropriately long time horizon
- Establishing an explicit timing mechanism for the initial investme
- Adopting a multiple factor approach and employing factors that diversify each other

Alpha vs Risk Factors

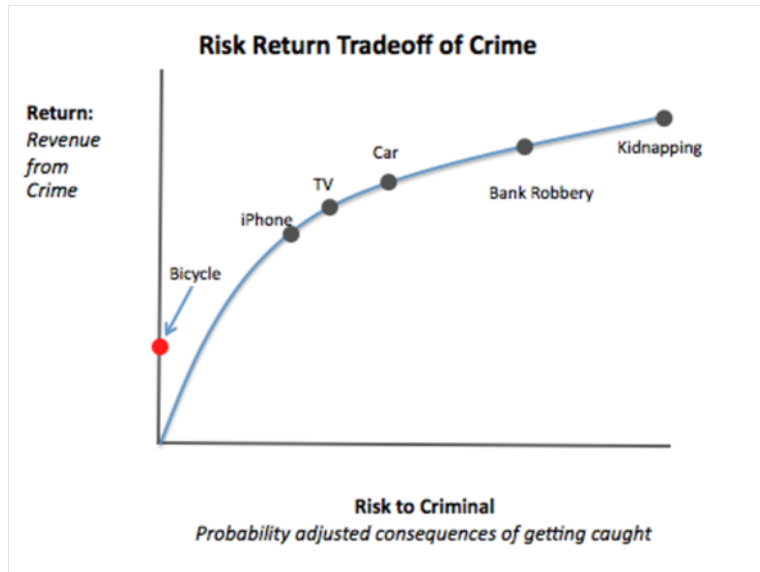
Exhibit 3: Differentiating Between Factors Reflecting Risk Premia and Other Factors (Cumulative Returns to Barra Global Equity Model (GEM2) Factors)



Why Factor Investing Works?

- In general, there are two main camps in the debate over what drives factor returns—one based on the view that markets are efficient and that factors reflect “systematic” sources of risk, and one based on the view that investors either exhibit behavioral biases or are subject to different constraints (e.g., time horizons, ability to use leverage, etc.).
- In the first camp:
 - The term “systematic” refers to the fact that risks to these stock traits cannot be diversified away.
 - This argument is consistent with “efficient markets theory” which assumes markets are efficient and investors are rational.
 - Here, factors earn excess returns because there is “systematic risk” attached to them.
 - For example, some have argued that the small cap premium is return earned for exposure to companies that are less liquid, less transparent, and more likely to be distressed.
 - Others have argued that factors like Value, Size, and Momentum are linked to important macroeconomic factors such as growth and inflation and because of their sensitivity to shocks in the economy, must bear a return premium.

Why Factor Investing Works?



Why Factor Investing Works?

- In the second camp:
 - Factors are thought to earn excess returns because of investors' "systematic errors."
 - If enough investors exhibit these biases, as long as it is prohibitively costly for rational investors to arbitrage these biases away, it can lead to the factor anomalies we observe.
 - Behavioral biases due to cognitive or emotional weaknesses:
 - Chasing winners, over-reacting, overconfidence, preferring "familiar" investments such as securities of the companies they work for or the country they live in ("home bias"), and myopic loss aversion.
 - Constraints and frictions/flows that arise from regulatory and industry practice. Here anomalies can arise from investors behaving rationally but subject to constraints.
 - Stocks with low liquidity earn a premium over long horizons (10 years plus) since most investors have shorter horizons (3- and 5-year horizons).
 - The use of institutional benchmarks, and the subsequent preference for relative returns, as one reason why the low volatility premium exists.
 - Reputation concerns cause managers to herd, and this generates momentum under certain circumstances.

Why Factor Investing Works?



Why Factor Investing Works?

	Risk-Based Explanation	Behavioural Explanation
Value	Costly reversibility of assets in place leads to high sensitivity to economic shocks in bad times	Overreaction to bad news and extrapolation of the recent past leads to subsequent return reversal
Momentum	High expected growth firms are more sensitive to shocks to expected growth	Investor overconfidence and self-attribution bias leads to returns continuation in the short term
Low Risk	Liquidity-constrained investors hold leveraged positions in low-risk assets which they may have to sell in bad times when liquidity constraints become binding	Disagreement of investors about high-risk stocks leads to overpricing in the presence of short sales constraints.
Size	Low profitability leads to high distress risk and downside risk. Low liquidity and high cost of investment needs to be compensated by higher returns.	N.A.

Source: ERI Scientific Beta (2014)

Why Factor Investing Works?

Exhibit 5: Theories Behind the Excess Returns to Systematic Factors

Systematic Factors	Systematic Risk-based Theories ¹⁷	Systematic Errors-based Theories ¹⁸
Value	➤ Higher systematic (business cycle) risk	➤ Errors-in-expectations ➤ Loss aversion ➤ Investment-flows-based theory
Low Size (Small Cap)	➤ Higher systematic (business cycle) risk ➤ Proxy for other types of systematic risk	➤ Errors-in-expectations
Momentum	➤ Higher systematic (business cycle) risk ➤ Higher systematic tail risk	➤ Underreaction and overreaction ➤ Investment-flows-based theory
Low Volatility	➤ N/A	➤ Lottery effect ➤ Overconfidence effect ➤ Leverage aversion
Dividend Yield	➤ Higher systematic (business cycle) risk	➤ Errors-in-expectations
Quality	➤ N/A	➤ Errors-in-expectations ¹⁹

Popularity of Factor Investing

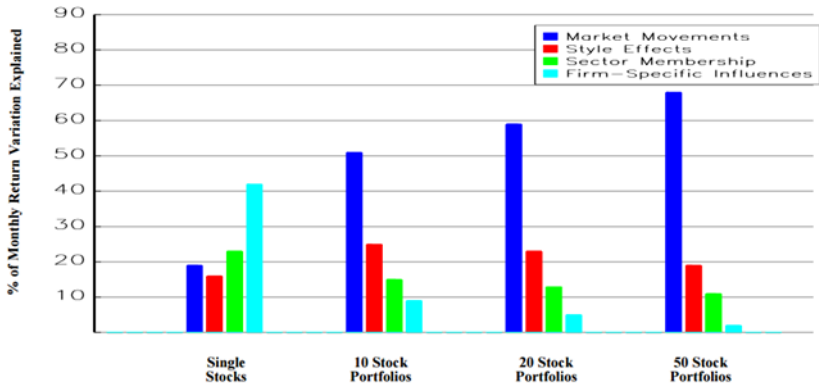
Factor investing

- **What?:** Factor investing is an attempt to capture systematic risk premia attached to certain alpha factors.
- **How?:** Portfolios are tilted towards securities that have high exposure to alpha factors.
- Factors and factor-based models form the core of a major part of today's quantitative trading strategies.

Active vs Passive Investing

Invest as a whole

Percentage of Monthly Returns Explained by Systematic Factors





US Factor Reference Book

What Drives Equity Returns?



North America Equity Research
20th January 2011

- In this analysis we investigate how our library of commonly used quantitative factors has performed when tested across the Russell 2000 equity universe, focusing on size, style, sectors, as well as different factor investment horizons.
- Given the significant breadth of US equity markets, it is clear that factor effectiveness does vary according to size. It comes as no surprise that, overall, factors have shown to be more effective at identifying asset returns of winners from last year's losers as the market capitalization of the universe they are tested on gets smaller.
- A number of factors were found to track well, however, in a long term valuation framework of metrics was most effective, in particular cash flow based factors. In addition to yielding strong long-horizon return profiles, information ratios, information coefficients, hit rates, and relatively high alpha decay and turnover, the theoretical factor returns exhibited statistical significance.
- Factors pertaining to growth and outlier/fundamentals, such as forward earnings momentum and analyst revisions, have shown greater forecasting ability, in particular within the small-cap space.

Global Quantitative Strategy
US Stock Strategies
Dominant Factor Driven
Dimitris Lefas-Dryguez
(1) 212 633 6000
dimitris.lefas@jpmorgan.com

AmeriGill Stock Strategies
Steve Hahn
(212) 260-6949
steve.hahn@jpmorgan.com

JP Morgan Securities LLC
New York, NY

Robert Smith
(212) 260-6949
robert.smith@jpmorgan.com

JP Morgan Securities Asia Pacific
Europe Stock Strategies
Marina Diaz
(34) 91 723 7447
marina.diaz@jpmorgan.com

JP Morgan Securities Ltd
London, UK

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What do quants do?

- Identify fundamental return-driving factors
- Create portfolios with optimal exposures to those factors
- Identify and hedge exposures to risk factors
- Use optimizers to make trade-offs
 - Seek to maximize expected returns, net of expected trading costs, subject to a risk budget

Global Equity Opportunities, %

Region Factor	% Exposure
America	0.40
Innovation	0.85
Materials	0.95
Technology	0.80
Europe	0.75
Emerging Equity	0.70
Management	0.10
Japan	0.05

Size Factors

Size Factor	% Exposure
Size	0.05
Value	0.15
Quality	0.10
Growth	0.10

Global Equity Opportunities by Sector Factors

Sector Factor	Global exposure	Emerging exposure
Consumer	0.20	0.25
Healthcare	0.10	0.15
Financial	0.15	0.20
Energy	0.10	0.15
Technology	0.15	0.20
Industrials	0.10	0.15
Consumer Services	0.10	0.15
Utilities	0.05	0.10

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Quantitative Research

- Research Analyst
- Trading & Sales
- Quantitative Analyst
- Investment Manager

A Disciplined Approach

A Three-Step Process for Alpha Generation

The Credit Suisse generates ideas around a three-step process for portfolio construction. (See Exhibit 1.) Innovation arises from the market, select stocks, and analyze their portfolio performance. This is accomplished using a suite of innovative tools that we have developed. Each step of our comprehensive process incorporates top alpha generators, essentially fundamental, well-known indicators to evaluate asset performance, and the market.

Comprehensive Set of Quantitative Products and Services

We publish research exploring topics, such as 1000s strategies, factor analysis, and the impact of macroeconomic trends. Our weekly and monthly tools aid managers in digesting market news and formulating future strategies. Our Research team also offers to clients a variety of research products. We provide extensive customer support for clients, such as investment strategy evaluation, ongoing testing, and direct assistance to clients establishing their own quantitative platform.

Exhibit 1. Credit Suisse's Core Quantitative Process for Alpha Generation

Market / Factor Analysis

- What Works?
- Short Interest / Volatility & Abnormal Returns
- Robust Evidence

Quantitative Equity Research

- Portfolio Performance Attribution
- Decisions on Active Weights

Overview

- We use 13 factor groups (styles) or 44 single factors to define individual style strategies (input)
- Generate ranking of sectors and countries (output)
- Sector ranking also calculated for three sub-regions of GEM, including Emerging Asia, CEEMEA and Latin America

The motivation for a GEM style strategy tool

There remains uncertainty about the global economy's recovery path and the probability of a double dip in the US economy, as well as persistent concerns on the European debt crisis weighing on the market. For the Eurozone, there is a generally agreed need for emerging markets recovery will continue to play a more important role on the global stage.

- Sector ranking is offered only within EMERG (4.0/2.0) sectors as there are not enough markets in Latin America and CEEMEA to calculate meaningful rankings for 30/40/2 (3.0/2.0) sectors.
- Country ranking is accordance with the style strategy inputs, for the first time, integrated into the tool to support an auto-diagnosis of emerging and transitioning the region.

For the generalist to understand the global style strategy tool, please refer to the report [Global Economic Outlook - Global Economic Outlook](#) (Boston, Czech - January 10, 16 January 2012).

Rise of Factor Investing

J.P.Morgan

Global Emerging Markets Equity
Research
06 June 2015

Q-Strategy Help in EM

Stock screens that combine thematic macro and quant

- MSCI EM is down 5% in the past month, underperforming DM. In line with the euro and Yen, EM currencies are weakening versus the US dollar. There is concern that EM central banks, influenced by the expected Fed lift-off, are not cutting rates fast enough. This correction offers a good opportunity to add to our OW markets, in our view, especially India (see Table 3).
- China stocks continue to score high on quant metrics (see Table 2). We are OW on China. MSCI China's EPS correlation to the troubled parts of the economy is low. Policy makers are active first to reduce the interest burden. The robust

Emerging Markets Equity Strategy
Sanaya Tavaris #C
(1-212) 622-6470
sanaya.tavaris@jpmorgan.com
J.P. Morgan Securities LLC

Adrian Mowat
(852) 2930-5510
adrian.mowat@jpmorgan.com

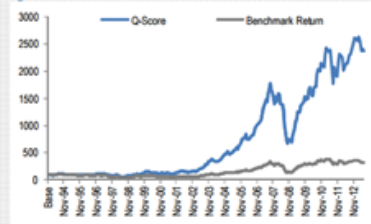
Is the Q-Score effective in GEM?

We can test the effectiveness of a stock-picking strategy by running a back-test.

Our GEM back-test rewinds the clock back to 1993 and examines the performance of those stocks with a Q-Score > 80% from then until now. (Scores are recalculated monthly and the portfolios are rebalanced monthly). The chart below shows the performance of these stocks versus an equal weighted benchmark.

Over this 17-year test period the top-ranking stocks have delivered an annualized return of around 18.5% converting an initial \$100 investment in 1993 into \$2270 in 2012 (excluding t-costs). This compares with a \$100 investment in the index which would have delivered a

Figure 1: Portfolio Index Performance



Source: J.P. Morgan Quant team

Rise of Factor Investing



PRESS RELEASE

BlackRock Appoints Dr. Andrew Ang to Lead Factor Investing Platform

By

Published: June 9, 2015 4:00 p.m. ET



Author and Renowned Systematic Investment Expert to Lead Team Managing \$125 Billion in Client Assets

NEW YORK, Jun 09, 2015 (BUSINESS WIRE) – BlackRock, Inc. **BLK**, **0.95%** announces the appointment of Andrew Ang, PhD as a Managing Director and head of the Factor-Based Strategies Group. In this role, Dr. Ang will lead BlackRock's expansion in the emerging field of active investing via exposure to different risk premiums. BlackRock already manages over \$125 billion in client assets across a variety of factor-based products and strategies.

Dr. Ang joins BlackRock from Columbia Business School, where he has focused on understanding the nature of risk and return in asset prices; particular the behavior of factor risk premiums within and across asset classes, over the past 15 years. His research spans bond market equities, asset management and portfolio allocation, and alternative investments. Most recently, Dr. Ang



David G. Pagano, CFA, Joel M. Dickson, PhD

- Factor-based investing is a framework that integrates factor-exposure decisions into the portfolio construction process.
- In this paper, we review, discuss, and analyze factor-based investing, drawing the following conclusions:

—Factor-based investing can approximate, and in some cases replicate, the risk exposures of a range of active investments, including manager- and index-based strategies.

INVESTMENT WEEK

investmentweek.co.uk April 2015



PIMCO

Your Global Investment Authority

Traditional portfolio construction approaches, which focus on asset class diversification, may fall short of investors' goals. A more efficient diversification strategy may be to allocate across the underlying "risk factors."

Understanding risk factor diversification

by William Bernstein

EDUCATION SERIES

Why isn't traditional asset class diversification enough?

Traditional allocation strategies seek to mitigate overall portfolio volatility by combining asset classes with low correlations to each other, meaning that they tend not to move in the same direction at the same time. However, asset class correlations are less stable than many investors realize, and long-term trends such as globalization are driving correlations higher. In addition, correlations typically increase during periods of market turbulence. As a result, seemingly distinct asset classes are likely to behave more similarly than many people expect.

In other words, even portfolios that are well diversified across asset classes may not be positioned to adequately diversify and cushion market volatility (Figure 1).

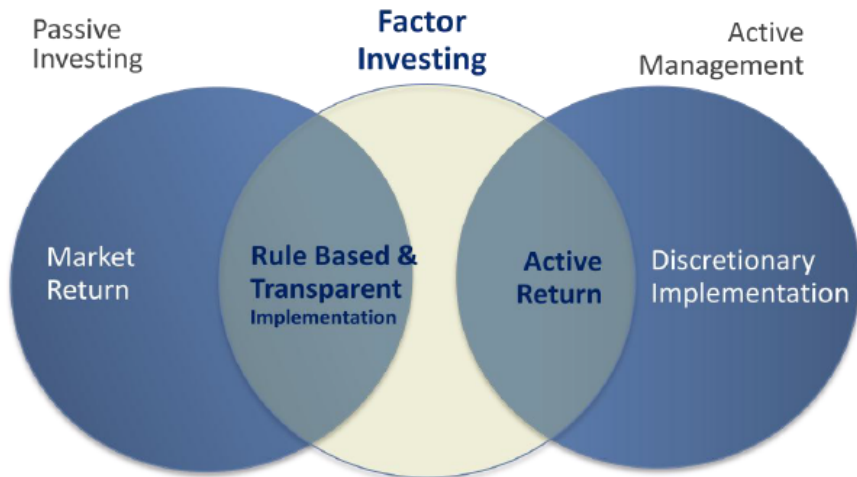
What is a risk factor?

Risk factors are the underlying risk exposures that drive the return of an asset class (see Figure 2). For example, a stock's return can be broken down into equity market risk – movement within the broad equity market – and company-specific risk. A bond's return may be explained by interest rate risk – prior sensitivity to changes in rates – and issuer-specific risk. And currency risk is a factor for non-U.S. dollar-denominated assets. By targeting exposure to these underlying risk factors, investors can select a mix of asset classes that provides more diversified portfolio risk.

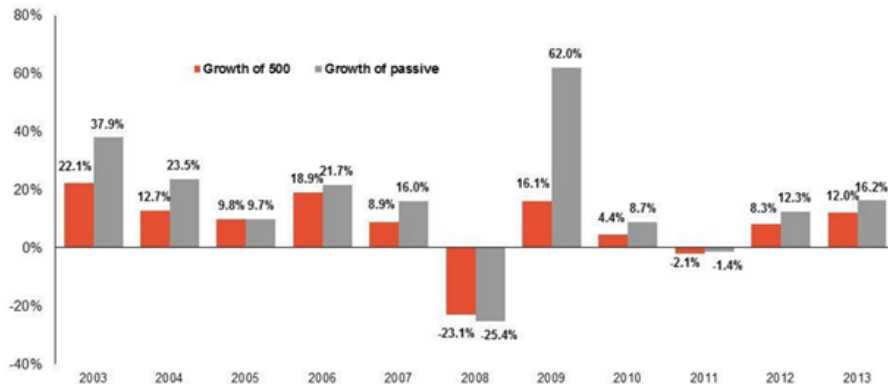
Figure 1: Asset class diversification is not the same as risk diversification

In the chart below, a portfolio is broadly diversified across asset classes, but in fact has a very concentrated exposure to underlying equity risk. Understanding these risk factors

Active vs Passive Investing



Active vs Passive Investing



CAGR of 500 since 2003: 5.8%
CAGR of passive since 2003: 12.4%

Source: Tower Watson (2014)

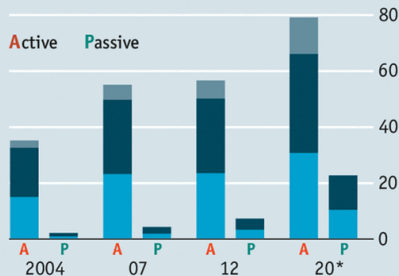
Active vs Passive Investing

Passive aggressive

1

Global assets under management, \$trn

■ Retail
■ Institutional
■ Hedge funds and private equity



Source: PwC

*Forecast

Rise of Factor Investing

MSCI

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MSCI FAMILY OF FACTOR INDEXES

SYSTEMATIC FACTORS	MSCI INDEXES
Quality	MSCI Quality Indexes represent the performance of high quality stocks by weighting based on debt-to-equity, return on equity and earnings variability. They are designed to reflect the performance of a quality growth investment strategy. High Exposure: MSCI Quality Index, MSCI Sector Neutral Quality Index High Capacity: MSCI Quality TR Index
Value	MSCI Value Weighted Indexes represent the performance of the value factor by weighting according to four fundamental variables (Sales, Earnings, Cash Flow, Book Value). High Exposure: MSCI Enhanced Value Index High Capacity: MSCI Value Weighted Index
Size	MSCI Equal Weighted Indexes represent the performance of the low size effect by equally weighting all stocks in a given parent index. High Exposure: MSCI Equal Weighted Index

S&P DOW JONES INDICES

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STRATEGY

ALL SINGLE-FACTOR MULTI-FACTOR RISK CONTROL DIVIDEND INCOME YIELD ASSET ALLOCATION MORE

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SINGLE-FACTOR

S&P Dow Jones Indices provides ways in which investors can gain exposure to a certain factor - or SR - such as volatility or beta.

ENHANCED VALUE FILTER FOR

ALL INDICES

S&P ENHANCED VALUE ENHANCED VAL

7,375.73 -61.96 -0.83%
PRICE RETURN DAILY CHANGE

12 MONTH

S&P ENHANCED VALUE ENHANCED VAL

2,946.11 -0.66 -0.02%
PRICE RETURN DAILY CHANGE

12 MONTH

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COMPLIMENTARY LIVE WEBINAR
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FTSE Global Factor Index Series

The FTSE Global Factor Index Series is a new suite of benchmarks designed to represent the performance of specific factor characteristics, with six single factor indices and factors comprising the index series. The six factors represent common factor characteristics for which there is a broad academic and practitioner consensus, supported by across different geographies and time periods.

- Liquidity
- Momentum
- Quality
- Size
- Value
- Volatility

The indices are based on the market cap weighted FTSE Developed and FTSE Emerging indices. The FTSE Global Factor Index series uses a transparent methodology to a target factor, whilst considering levels of diversification and capacity.

Index Resources

- Index Methodology Overview
- Index Construction Methodology
- Momentum Factor
- Quality Factor
- Value Factor

Fact sheets

- FTSE All World Factor Indices
- FTSE Developed Factor Indices
- FTSE Developed ex US Factor Indices
- FTSE Developed ex Korea Factor Indices
- FTSE Latin America Factor Indices
- FTSE USA Factor Indices

See All

Methodology

- FTSE Global Factor Index

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Russell High Efficiency™ Factor Indexes

Efficient and effective tools for factor-based investing

The Russell High Efficiency™ Factor Indexes employ a consistent, factor-based weighting methodology to deliver institutional quality targeted exposure to commonly identified factors:

- **Low Volatility**
Identifies companies with low volatility—historically showing more stable return patterns than the broader market
- **Momentum**
Focuses on stocks with strong performance over the past 12 months (excluding the most recent month).
- **Quality**
Selects stocks that historically have had a greater ability to deliver sustainable returns, typically due to high profitability, low leverage and low earnings volatility.
- **Value**
Identifies stocks that are currently trading at a discount to fair value.



Related resources

- Construction & methodology (PDF)
- Returns, values and ticker symbols

Rise of Factor Investing

iShares by BLACKROCK

iShares ETFs ETF Strategies About ETFs iShares Resources

Smart Beta

Think differently about investing, think smart beta

iShares Smart Beta ETFs

Exposures

Smart Beta exposures can help investors to tailor factor exposures and implement specific views.

Equity Factors

MTUM	VLUE	QUAL	SIZE	IGLT	MTM
Momentum	Value	Quality	Size	Int'l Quality	Int'l Momentum

Minimum Volatility

USMV	EFAV	EDMV	ACWV	JPMV	AXLV	EURV
US	EAFE	Emerging Markets	All Country	Japan	Asia ex-Japan	Europe

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Factor Investing

The Global Leader in Factor-Driven ETFs

PowerShares S&P Factor Driven ETFs provide new takes on classic indexes. These groundbreaking products have the potential to provide low volatility, high beta, high quality, and high dividend tilts on traditional domestic and international benchmarks.

In addition, the Factor Driven suite offers the benefits of the ETF structure, including intra-day liquidity,¹ transparency,² and tax efficiency.³

Low Volatility Momentum High Beta High Quality High Dividend Resources

Access Unconstrained Low Volatility Strategies from Invesco PowerShares

We believe managing the impact of volatility is critical when pursuing long-term investment goals — that's why PowerShares offers the industry's largest suite of S&P Low Volatility ETFs, so you can choose the solution that's right for your needs. Log-in to view a suite of ideas from PowerShares Global Product Strategy experts.

Rise of Factor Investing

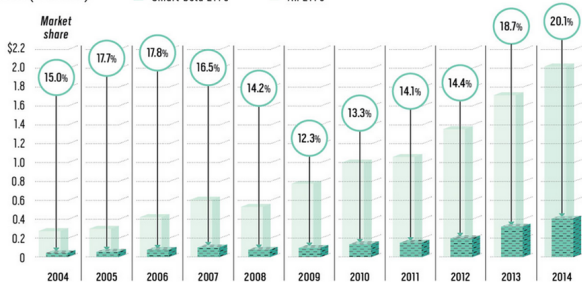
GROWTH SPURT

ETFs have grown into a category with more than \$2 trillion in U.S. assets. Smart-beta funds now account for one-fifth of that figure.

Assets (in trillions)

Smart-beta ETFs

All ETFs



Source: Bloomberg

Rise of Factor Investing

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FACE TO FACE

May 17, 2015 8:28 am

FTSE chief's smart-beta warning for active funds

Steve Johnson

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Active fund managers may wish to look away now: if Mark Makepeace is correct, so-called "smart beta" is about to eat your lunch.

According to the chief executive of FTSE Group, the index provider, traditional market capitalisation-weighted passive investment now accounts for 30 per cent of the money invested in major stock markets such as the US and UK, leaving active managers with 70 per cent.

But "half of that [70 per cent] over time will probably move to a combination of factor and multifactor-based [passive] approaches," Mr Makepeace says, with potentially devastating consequences for

active managers.

Factor-based investing, which might involve weighting companies by their revenues or book value, or creating equal-weighted portfolios, is a craze currently sweeping the investment firmament.

Dr. Ayhan Yuksel, CFA, FRM, PRM

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EM SQUARED

June 8, 2015 6:49 pm

Smart beta offers promise as Chinese markets open up

Steve Johnson

China A-shares displays low correlation with other regions

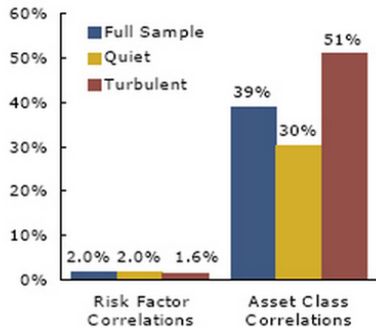
3-Year correlation (2012-14)	China A	China H	World	EM	USA	Europe	Japan	AC APxJ
China A	1							
China H	0.59	1						
World	-0.01	0.47	1					
EM	0.15	0.71	0.8	1				
USA	-0.03	0.36	0.94	0.68	1			
Europe	-0.01	0.45	0.94	0.74	0.8	1		
Japan	0.01	0.36	0.67	0.58	0.55	0.58	1	
Asia-Pacific ex-Japan	-0.14	0.74	0.78	0.97	0.64	0.72	0.58	1

Sources: SSGA; MSCI; FactSet, as of December 31 2014

But do these factors apply in a market that is 90 per cent-owned by domestic retail investors, where turnover is so high the average holding period is just two months, and where investors see nothing odd in stocks surging in value simply because they have switched between market sectors and an entire market (step forward the Shenzhen Composite) trades on a multiple of 62.7 times earnings?

Diversification Benefits

Average Cross-Correlations (March 1994 – December 2009)



Risk Factors		Asset Classes
Equity	EM Spread	US Equities (Small Cap)
Size	Mtge Spread	US Equities (Large Cap)
Value	Corp Spread	Global Equities
Momentum	Swap Spread	Emerging Market Equities
Duration	Real Estate	Bonds
2-10 Slope	Commodities	Real Estate
10-30 Slope		Commodities

Is Factor Investing a Hype?

HYPE ?